

Vocabulary

Translation: a slide of the graph — $f(x) + k$ moves it up/down, and $f(x - h)$ moves it left/right.

Reflection: a flip: $-f(x)$ flips over the x -axis, and $f(-x)$ flips over the y -axis.

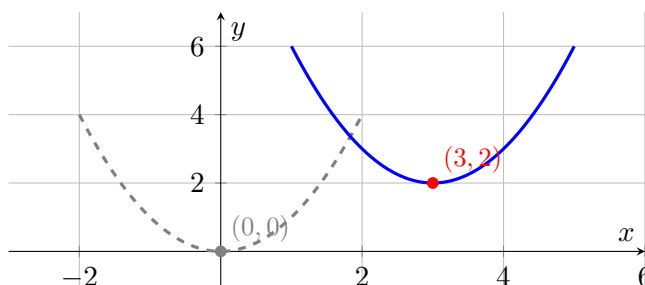
We do together. Describe how $g(x) = (x - 3)^2 + 2$ transforms the parent x^2 .

Step 1. Inside the square, $x - 3$ shifts the graph right 3 (opposite the sign).

Step 2. The $+2$ outside shifts it up 2.

Step 3. So the vertex moves from $(0, 0)$ to $(3, 2)$.

See it:



parent (dashed) → right 3, up 2 (solid).

You Try. Solve each. Show every step, then name the answer.

a) Describe $h(x) = -x^2$ from the parent.

reflection over the x -axis (opens down)

b) Give the vertex of $y = (x + 1)^2 - 4$.

$(-1, -4)$

Summary (fill in): $f(x - h) + k$ shifts a graph h right, k up; a leading $-f(x)$ reflects it over the x -axis.